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# Influence of Social Variables on the Biobehavioral Response to Separation in Rhesus Monkey Infants

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LEVINE, SEYMOUR; FRANKLIN, DEBORAH; and GONZALEZ, CAROL A. *Influence of Social Variables on the Biobehavioral Response to Separation in Rhesus Monkey Infants.* CHILD DEVELOPMENT, 1984, **55**, 1386–1393. Effects of social stimuli on behavioral and physiological responses to separation were examined in 4-month-old rhesus monkeys. Infants were removed from their social group under 3 counterbalanced conditions: (1) infant totally isolated from its familiar physical and social environment and placed alone in a cage for 4 days; (2) "mother in"—housing the infant in a single cage in front of its social group with mother remaining in the group; (3) "mother out"—similar situation, except that mother was removed from the group during separation. Infant behavior was recorded, and, at selected times during separation, a blood sample was obtained for analysis of plasma cortisol. Infants rarely vocalized when totally isolated, but showed high vocalization and movement in the presence of their social groups. Vocalization was transiently higher in mother-out condition than in mother-in condition. Infants never showed signs of depression. Plasma cortisol response did not differentiate between groups. Animals showed significantly high levels of plasma cortisol 3 hours following separation. These data indicate that the responses of the infant following separation are attempts to produce effective coping responses. The concepts of "protest" and "despair" are discussed as they relate to behavioral differences observed following different separation paradigms.

The behavioral sequelae following separation of the nonhuman primate mother and infant have been studied extensively (Hinde & Spencer-Booth, 1967; Kaufman, 1973; Mineka & Suomi, 1978; Seay, Hansen, & Harlow, 1962). The response of the infant following separation has been characterized as biphasic. This involves an initial "protest" response, involving increased levels of activity and vocalization, followed by a "depression" in which the infant shows decreased activity, a hunched-over posture, and a withdrawn state often involving self-directed behaviors.

The behavioral response by the primate infant to maternal separation does show considerable variation, and often the biphasic protest-despair response has not occurred (Lewis, Young, McKinney, & Kraemer, 1976; Rosenblum & Kaufman, 1968). Several factors appear to influence the separation response of the infants. These are the nature of the separation environment, preseparation social history, and species (Mineka & Suomi, 1978). Even among separations involving one species, rhesus monkeys (*Macaca mulatta*), there has been great variation in the behavioral responses to maternal separation, as well as in the inferred severity or stressfulness of the responses observed using different experimental procedures (Chappel & Meier, 1975; Hinde & McGinnis, 1977; Lewis et al., 1976; Seay & Harlow, 1965). In our laboratory, previous separations involving socially reared rhesus monkeys have failed to produce behavioral responses indicative of depression even after a prolonged (2-week) separation from their mothers (Gunnar, Gonzalez, Goodlin, & Levine, 1981).

One of the major concerns in several previous studies was the nature of the social environment of the infant, including whether separation from the mother was total or only partial and whether the infant was separated from its familiar peers into a novel environment. Seay and Harlow (1965) found that the behavioral responses of rhesus infants separated from their mothers by Plexiglas were more exaggerated than if their mothers were totally removed into another room. Addition-

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ally, Hinde and Davies (1972) reported that removing the infants from their group into a novel environment and housing them individually prolonged the display of protest behavior and delayed the onset of depression.

Physiological indexes of arousal, particularly the pituitary-adrenal response, have been shown to be sensitive to maternal separation in young squirrel monkeys (Coe, Mendoza, Smotherman, & Levine, 1978; Mendoza, Smotherman, Miner, Kaplan, & Levine, 1978) and rhesus monkeys (Smotherman, Hunt, McGinnis, & Levine, 1979). Rhesus monkey infants in our laboratory have shown significant elevations in plasma cortisol at 30 min and 3 hours after separation compared with nonseparated controls (Gunnar et al., 1981; Smotherman et al., 1979). The time course of the plasma cortisol elevation also paralleled that of the "whoo call" vocalization, which showed a concomitant increase and subsequent decline within 24 hours. Thus, in these studies it appeared that the pituitary-adrenal and the behavioral responses are correlated and sensitive indicators of the arousal induced by separating the infant from its mother.

Our study examined the behavioral and pituitary-adrenal responses of rhesus monkey infants when allowed visual and auditory (but not tactile) access to their familiar surroundings and social peers, and compared their responses to total isolation in a novel environment. The experimental conditions in this study closely resemble those studied by Seay and Harlow (1965) and Suomi, Collins, Harlow, and Ruppenthal (1976).

## Methods

*Subjects.*—The subjects were five rhesus monkeys approximately 4 months of age. All were born into two social groups that had been established at Stanford University for approximately 10 years. Each group consisted of one adult male and eight adult females and their juvenile offspring. Three of the infants (two females and one male) were derived from one of the social groups, and two of the infants (one female and one male) were obtained from the other social group. The groups were maintained in two outdoor enclosures (5 × 10 × 3 meters) separated by a chain-link fencing that adjoined smaller indoor quarters (3 × 5 × 5 meters).

*Procedure.*—Each of the subjects was studied under three different separation procedures. Each separation was 96 hours in duration, and the period between separations was 2–3 weeks. The length of time between separations was balanced across subjects. Thus, all subjects experienced equivalent interseparation periods. The order of separations was counterbalanced for each subject over a 10-week period. Only one type of separation for any single subject was performed during the 96-hour period. However, often two infants were simultaneously tested during the 96-hour period under different conditions. The conditions of separation were as follows: (1) "mother in" entailed housing the infant in a single cage (64 × 74 × 38 cm) in front of its social group with its mother remaining in the group; (2) "mother out" involved a similar situation to the first, except that the infant's mother was removed from the group during the separation; (3) the infant was totally *isolated* from its familiar physical and social environment and placed alone in a cage in a small room for the duration of the separation. When two infants were separated simultaneously, the first infant was in the outside condition; the second infant was in the isolated condition in which no visual or auditory contact with the social group was available. During this period of separation, the nonseparated subjects remained with their social group. All subjects were at one time in the social group home condition while others were in the captured condition, but the order was always balanced.

The experimental cage for the infants housed outdoors was secured 3 feet from the outdoor enclosure of the social group. The infant could observe both social groups; therefore, it was never possible to separate more than one infant at a time under the social group conditions. The infants were sheltered from the night with additional insulation, and heating was provided. A vitamin-fortified fruit drink was provided for all separated infants.

The subjects were captured at 1 P.M. on day 1, removed from their mothers and the social group, and placed into the experimental cages. Only one subject at a time was tested in conditions 1 or 2 (social group conditions) while the infants in the isolation condition were tested concurrently with the infants housed outdoors. Behavioral observations were made 90 min prior to obtaining blood samples. Blood samples were obtained at 3, 24, 48 and 96 hours after capture and separation.

The behavioral measures were scored onto a checksheet by an observer seated behind a screen viewing the subjects through reflective single-vision film. The categories

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of behavior were recorded within 30-sec intervals for 20 min. The behaviors scored were frequency of movements, vocalizations that included "whoo calls" (Gunnar et al., 1981), and screams. To maintain accuracy in the scoring of other ongoing behaviors, the vocalizations were scored as a frequency until a maximum of five per 30-sec interval was obtained. Since the movements of the infants were sometimes continuous throughout an entire 30-sec interval, continuous movements were scored as such and given a maximum frequency score of 5, which was summed with the discrete shorter duration movements. Stereotypy (the incidence of repetitive, nonchanging motor patterns) was also scored, as were eating and drinking. The infants' upright or hunched-over posture was recorded on-the-signal for any indications of the infants' lapsing into depression. Measures of proximity to the infant by group members and maternal agitation were recorded for 20-min periods.

The blood samples were obtained via saphenous venipuncture by restraining the unanesthetized infant. All samples (.4 ml) were then centrifuged, and the plasma was collected and frozen until radioimmunoassay. Plasma cortisol was assayed using antiserum F21-53 (Endocrine Systems, Tarzana, California) using the method of Klemm and Gupta (1975). This method has been previously described in detail in this laboratory by Smotherman et al. (1979).

## Results

The effect of the experimental conditions of separation for the rhesus infants was most marked in the vocalization responses. A repeated measures 3 (conditions) × 4 (time) ANOVA showed significant main effects for conditions, $F(2,12) = 28.80$, $p < .001$, and time, $F(3,36) = 11.02$, $p < .001$, for the frequency with which whoo calls were emitted. There was also a significant conditions × time interaction, $F(6,36) = 2.80$, $p < .02$. Tests for simple main effects and the Newman-Keuls post hoc tests revealed that the infants exhibited significantly more whoo calling when they were in the two conditions exposed to their social group as compared with being isolated at all times, $p$'s $< .01$. Additionally, the frequency of whoo calls was greatest on the first day shortly after capture and separation (3 hour) as compared with the behavior at 24, 72, and 96 hours later, $p$'s $< .05$, as shown in Figure 1. Furthermore, there was a more rapid decline in the incidence of whoo calls between 3 hours and 24 hours

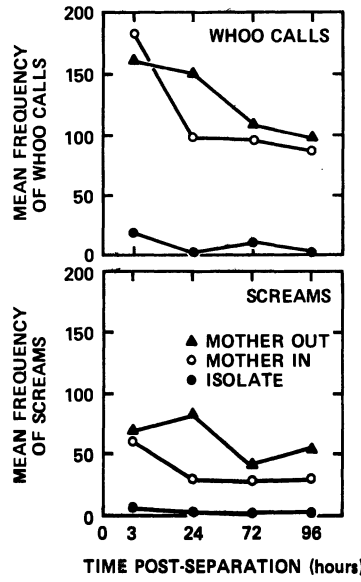



FIG. 1.—Mean vocalizations (whoo calls and screams) per 20-min observation following separation.

following separation when the mother was present ("mother in") than when she was absent from the group ("mother out"), thus accounting for the significant conditions × time interaction, $p < .05$. The number of screams was significantly greater when the infants were in view of their familiar groups as compared with the isolate condition (Figure 1). As with whoo calls, there was a significant main effect of conditions, $F(2,12) = 32.40$, $p < .001$. Post hoc tests showed that the two social group conditions, mother out and mother in, were significantly greater than the isolate at all times, $p$'s $< .05$. The significant conditions × time interaction for screams, $F(6,36) = 3.01$, $p < .02$, was also because of a greater number of screams emitted by the infants when their mother was out of the social group than when she was present 24 hours after separation.

The frequency of movements showed similar differences between the mother-out and mother-in conditions in which the infant viewed its social group and when housed in isolation, but they were not as discrepant as the vocalizations (Figure 2). There was a significant main effect of conditions, $F(2,12) = 14.20$, $p < .01$. There was also a significant conditions × time interaction for the movements. Post hoc Newman-Keuls tests indicated that there was a significant difference between the mother-out and mother-in conditions 24 hours after separation, $p < .01$. Both of the social group conditions differed

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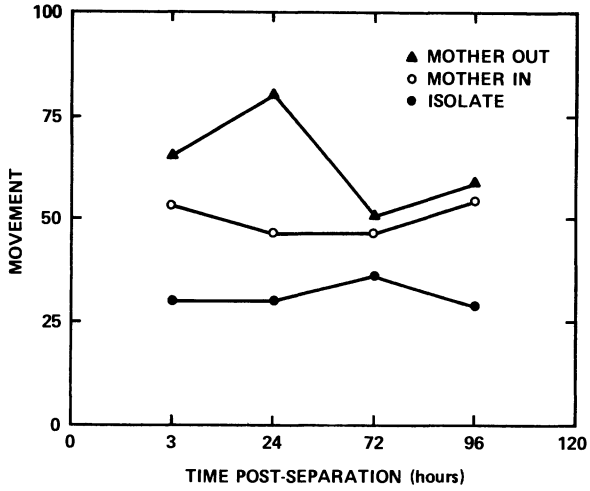



FIG. 2.—Mean frequency of movements per 20-min observation period. Movement is defined as the number of movements per 30-sec interval with a maximum score of 5 assigned to any 30-sec interval where continuous movement occurred.

significantly from the amount of movement observed during the isolate condition, $p$'s < .05, except at 72 hours after separation.

Other behaviors, such as stereotypies, occurred so infrequently that a parametric test could not be used. However, stereotypies were observed when the infants were in front of their social group during the initial 24 hours and only when their mother was present in the group. The infants continued to eat and drink throughout the durations of all the separations.

The plasma cortisol levels did not discriminate between the three experimental conditions, $F(2,12) = .65$, $p < .74$, nor was there a significant conditions × time interaction, $F(6,36) = 1.13$, $p < .36$. There was, however, a main effect of time, $F(3,36) = 16.72$, $p < .001$. Tests for simple main effects indicated that the cortisol levels in the infants 3 hours after separation were significantly greater than those measured at 24, 72, and 96 hours, $p < .05$. The cortisol levels over the duration of the separation are shown in Figure 3, in which the plasma cortisol levels (μg/100 ml) are combined over all the experimental conditions.

## Discussion

The findings in this study clearly demonstrate that the behavioral sequelae that have been reported following disruptions of the mother-infant relationship depend on the environmental conditions prevalent at the time of separation. Thus, providing separated infants with visual and auditory access to, but preventing the ability to interact with, the mothers and social peers elicited far more vocalization and movements than complete isolation of the infants in a novel environment. The fact that the behavioral responses to separation are variable does not come as a surprise. It has already been pointed out that one potential source of variance is the differences in the mother-infant separation paradigm (Mineka & Suomi, 1978). However, the surprising aspects of these data are the relatively minimal number of vocalizations produced by animals that are socially isolated. These data have been replicated in a subsequent study with rhesus monkeys (Johnson, Gonzalez, & Levine, 1982). Per-

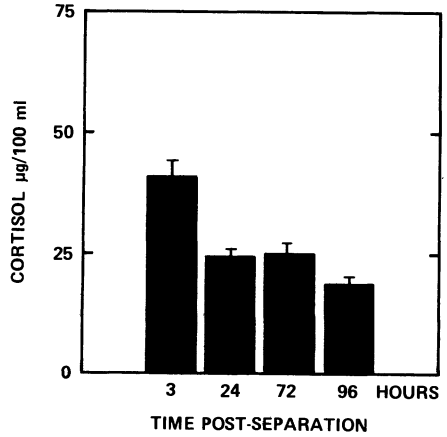



FIG. 3.—Mean plasma cortisol (±SE) of the separated infants at different times following separation. All conditions are represented in the mean values insofar as experimental treatments were not a discriminating variable.

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haps one of the reasons that this phenomenon has not been previously noted is that most separation paradigms do not involve total isolation. Either the infant is separated in the presence of other isolated infants or cues emanating from other animals have not been rigorously controlled for. We should emphatically point out that, even though the socially isolated infants do not vocalize extensively, there is no evidence that these animals are in what has been characterized as "depression." Neither typical postures nor lack of movement was observed in these animals. They appear to be active and exploratory, and the eating and drinking behavior was normal.

There have now been a number of reports that appear to indicate that the behavioral response to separation does have environmental constraints. In certain species of animals, removal of the mother from the social group does not elicit the vigorous "protest" response when the infant is permitted to remain in the social group. Thus, squirrel monkeys (Coe et al., 1978) and bonnet macaques (Rosenblum & Kaufman, 1967) appear to show little behavioral disturbance by removal of the mother when social partners are available. It was hypothesized that the bonnet macaques fail to show distress following removal of the mother because of the social network available to the infant, in that there were available aunts who had previously participated in maternal care. In some ways, the squirrel monkey has a similar social organization to the bonnet macaque, in that aunting is frequent. Aunting usually refers to the participation of a female other than the mother in caretaking behavior of the infant. Although lactation is not a frequent aspect of this behavior, the carrying, retrieving, and transport of the infant is often indistinguishable from that of the maternal mother.

In a recent article by Suomi, Mineka, and DeLizio (1983) there appears this statement: "There is now considerable evidence that when most human and nonhuman primate infants are forcibly separated from their mothers, they quickly become agitated and distressed, a reaction first termed 'protest' by Robertson and Bowlby (1952). Most researchers studying this phenomenon have considered protest to be a universal initial reaction to maternal separation for primate infants, probably the product of natural selection (Kaufman & Rosenblum, 1967; Rosenblum & Plimpton, 1981)." The data in our study are certainly at variance with this position. The so-called protest reaction does not appear to be invariant and also appears to be subject to environmental constraints, as are many other behavioral parameters that have been studied in relationship to separation. In fact, these data create problems in attempting to resolve the whole notion of the term "protest." "Protest" has been defined, in primate species at least, as that initial period following separation that is characterized by intense vocalization and agitated movements. In the experiment by Hinde and Davies (1972), where infants were removed from their social group and placed in a strange environment, it was reported that the separated infants in a strange cage seemed to be even more disturbed (vocalized more) than infants that remained in the home cage while the mother was removed. It is important to note, however, that, in this study, although the infants were removed to a strange cage, these infants were in auditory contact with either the social group or other separated infants.

Although the concept of protest does have its operational definition in terms of vocalization and movement, the term clearly implies a set of affective responses that are above and beyond its operational definition. The term elicits a complex set of surplus meanings that certainly extend beyond its operational usage. It is possible to entertain a completely different set of hypotheses regarding vocalizations within the context of this experiment that have little to do with the concept of protest and that deal specifically with the adaptive value of the vocalization response following separation and the stimuli that would be capable of eliciting it. We have attempted to deal with certain aspects of the mother-infant relationship and, in particular, the behavioral events that occur under separation, within the context of coping theory (Levine, 1980). We propose that, under certain conditions of stress, proximity and contact with the mother results in the modulation and reduction of the infant's arousal levels. Since coping requires control as a mechanism for reducing stress, proximity-seeking behavior that results in arousal-reduction can best be viewed within the context of coping. Clearly, one of the major aspects of proximity-seeking behavior in the infant primate is vocalization. Vocalization in the primate social group generally results in the mother rapidly achieving proximity with her infant and frequently retrieving that infant. Therefore, the infant uses vocalization as a signal to achieve contact and also as a mechanism whereby it can control certain aspects of the mother's behavior. Rosenblum and Plimpton (1981) have argued

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that one possible function of the higher rates of vocalization observed during the first day of separation may be an attempt on the part of the infant to reestablish contact with the mother. The data from this study would support the view that vocalization is a signal that would, under normal circumstances, result in achieving proximity to the mother. If vocalization does represent the effective response that would achieve proximity and thus reduce arousal levels, then the absence of the stimuli to elicit such a response should result in a diminution of vocalization insofar as the relevant cues necessary to elicit this response are no longer present. If, however, the stimuli are available, then one would predict that vocalization should be emitted at an initially high rate and then tend to show some diminution as the response is not reinforced. The data from this study support both of these predictions. In the absence of the eliciting social stimuli, vocalizations are markedly diminished. Even within the group in which social stimuli are available, although vocalization continues at a significantly higher rate than the socially isolated condition, they do show a significant diminution over time.

There is another aspect of this data that can be explained within the context of coping theory. Although the conditions where social stimuli are present do differ dramatically from those in which the animals are separated and socially isolated, even within the social stimuli conditions there was a transient but significant effect of whether the mother was present or absent from the social group during separation. Thus, we find that, in both vocalizations and activity, there is a transient but significant increase in both of these behavioral parameters if the mother is absent. One could argue that the increased vocalization in the absence of the mother represents a continued attempt on the part of the infant to reestablish proximity and that the mother's presence leads to a more rapid extinction of this response, since the predicted response of the mother—namely, attaining proximity—no longer occurs when they are physically separated.

Although there are clear behavioral differences between the two conditions of separation, the physiological response to separation—changes in circulating levels of plasma cortisol—does not discriminate between these conditions. There are a number of factors that could have contributed to this outcome. First, within the context of this experiment, animals were removed from the social group and immediately isolated. Thus, in addition to the consequences of separation that have been shown to influence pituitary-adrenal activity, there is the response to capture, which is itself a potent stress (Gonzalez, Gunnar, & Levine, 1981), in addition to now being placed in a novel environment. It is not possible to determine which of these stimuli could have been responsible for all of the groups showing similar cortisol elevations. The data do, however, address one issue, which is that in a prior study (Gunnar, Gonzalez, & Levine, 1980) it was reported that there was a positive correlation between vocalization and plasma cortisol responses following separation. These data do not support such a correlation. The marked differences in the behavior that we have discussed are not evident in the plasma cortisol values. It does appear that all animals tend to show an immediate response followed by a decrement that stabilizes over the next 96 hours. However, it should be noted that all of these values are significantly higher than we have obtained in a subsequent study using rhesus macaques in which basal levels were obtained in the late afternoon. These basal levels were approximately half those that we have observed in these animals when sampled at the same time. Furthermore, in a subsequent study (Johnson et al., 1982), when the effects of novelty (capture) are controlled for, the infant's cortisol response does discriminate between the presence or absence of the mother in the separation environment, but in this case the mother's absence stimulated a significant cortisol elevation not found when she was present. Thus, an inverse relationship between vocalization and pituitary-adrenal activity was observed. A similar finding has been reported for the squirrel monkey (Levine, Wiener, & Coe, 1982).

Finally, it should be noted that, in general, the separation response is described as being biphasic, with an initial "protest reaction" followed by despair. "Despair" is normally defined as an absence of vocalization and a diminution of agitated movement (Hinde & Davies, 1972; Mineka & Suomi, 1978). This is in contrast to the depression reaction (Kaufman & Rosenblum, 1967; Mineka & Suomi, 1978; Reite, Short, & Seiler, 1978), which is described as total inhibition of movement, including huddled-over posture. In terms of these definitions, we would have to reach the conclusion that our socially isolated animals do not have a protest response, but rather are in a state of despair. In contrast, we would have to argue that animals who are isolated in the presence of

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social stimuli show the initial protest response followed by despair, but that the despair is milder, since they are showing continually higher vocalizations. We feel that neither of these concepts are adequately descriptive of the data we have obtained. There was no quantitative or qualitative evidence that despair was present in these animals. The use of coping theory as an explanatory system more adequately deals with the data obtained in this experiment than do the more prevalent concepts that have pervaded the study of mother-infant relationships and, in particular, separation in primates. When one becomes steeped in the literature concerning separation experiences, it is easy to come to the conclusion that the infant is a delicate and vulnerable organism that has little resources to deal with these traumatic experiences. We would rather argue that the primate infant is indeed an extremely resilient organism that initially actively attempts to cope with the stress that is invoked by experiences related to brief periods of maternal loss.

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